

Draft - expert review pending

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Topic 4.5 - Principles, measures and models of HSE management

A real classroom situation

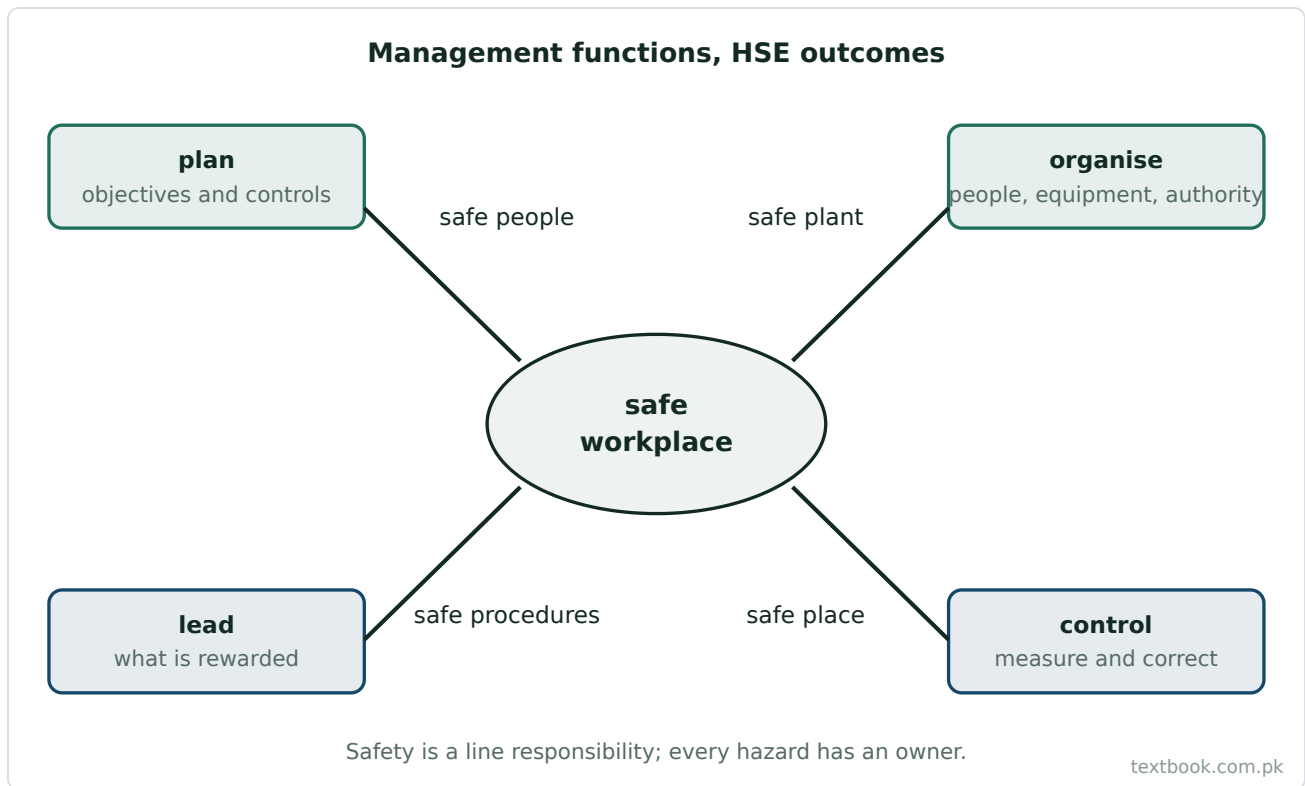
A student asks the safety officer from topic 4.3's workshop why he keeps the near-miss book at all. "Nothing in it is an accident," she says. He shows her the page: a dropped spanner, a slipped ladder, a smoke smell. "The pyramid says the accidents live under these," he says. "Count the small ones and you have counted the big one's shadow."

This topic is the theory behind his notebook: the principles that connect management to safety, the measures that make it visible, and the models that explain why small things are big things in disguise.

Explanation

Key principles of management and HSE; measures and models

Management runs on four functions, plan, organise, lead, control, and HSE borrows them wholesale. Applied to safety: **plan** the objectives and controls (Topic 4.2's PDCA), **organise** people, equipment and authority around them, **lead** by what is rewarded (Topic 4.3's philosophy layer), and **control** by measuring and correcting. The principles that tie the functions to HSE are old and reliable: safety is a line responsibility, not a staff department; every hazard gets an owner; and what is measured is what gets managed.



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The **measures** are the indicators of Topic 4.2, led by the leading kind, plus audits and safety-climate surveys that ask the workers what management's philosophy actually is. The **models** give the measures their meaning.

- The incident pyramid: for every serious injury, a large number of minor injuries and a very much larger number of near-misses and unsafe acts sit beneath it. The exact ratios are debated; the shape is not. The spanner and the smoke smell are the base of a pyramid whose apex is the fatality.
- The Swiss cheese model of accident causation: defences are slices of cheese with holes, and harm passes through when the holes line up. The Karachi fire's locked door, missing alarm, stored combustibles and unaudited housekeeping were four holes lining up. Every model of control is an attempt to keep the holes from aligning.
- Near-miss reporting, formalised, is the cheapest measure in the set: the pyramid's base is where the next apex is visible.

Three HSE measures and models

Model or measure	What it shows	What it is silent about
Incident pyramid	serious harm sits on a wide base of near-misses	the exact ratios (debated)
Swiss cheese model	defences fail when their holes line up	which hole moves next
Safety climate survey	what workers believe management rewards	what management says it rewards

Near-miss reporting is the cheapest measure: the pyramid's base is where the next apex is visible.

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Organizational environment

The organizational environment is the workplace's own climate: how work is actually divided, how pressure is applied, how bad news travels. It is the medium the models live in. A mill where the near-miss book is read has a different organisational environment from one where it is used to blame, and the two will produce different pyramids from identical machinery. The environment is measurable, by the questions workers answer about who gets punished for reporting, and it is manageable, by the same philosophy layer as ever.

A worked example. Read the uncle's mill through the models. The near-misses, the dropped bale, the unreported cough, are the pyramid's base. The defences, the guard, the mask rule, the annual medical, are cheese slices with holes (the guard off for convenience, the mask rule unenforced). The near-miss book, if the officer from 4.4 opens one, is the measure that shows both. Nothing new is needed except the reading.

Common misconception. "The pyramid's ratios mean small accidents cause big ones." Near-misses do not cause fatalities; they share causes with them.

That is the point: the base and the apex grow from the same soil, so gardening the base, fixing the small failures, shrinks the apex you never see coming.

Activity: Find the holes

Work in pairs. Your teacher gives you the Karachi fire's four failures as cards: locked exits, no alarm, stored combustibles, no housekeeping audit. Draw four cheese slices, place one hole per slice, and show the arrow of harm passing through. Then add a fifth slice of your own design that would have stopped the arrow, and name which control level it sits at. Allow 20 minutes.

Check your understanding

1. Name the four management functions and match each to one HSE action. *(Remember)*
2. What does the shape of the incident pyramid tell a safety officer to count, and why? *(Understand)*
3. The Swiss cheese model calls defences "slices with holes". Explain the Karachi fire in those terms. *(Analyze)*
4. A near-miss book exists but is empty. Is that good news? Use the pyramid and the organizational environment in your answer. *(Analyze)*

Summary

HSE management borrows management's four functions, plan, organise, lead, control, and binds them to safety by principles: safety is a line responsibility, every hazard has an owner, and what is measured is managed. The measures are led by leading indicators and near-miss reporting; the models, the incident pyramid and the Swiss cheese model, explain why the small and the serious share causes and why defences fail by alignment. The organizational environment, how pressure and bad news actually travel, is the medium that decides whether the models are used or ignored.

Self-assessment checklist

Sign in to sync your ticks across devices.

- ☐ I can match the four management functions to HSE actions.
- ☐ I can explain the pyramid's shape and what to count because of it.
- ☐ I can redraw a real case as cheese slices and add a working slice.
- ☐ I can explain why an empty near-miss book is not good news.

Try this at your practicum school

Start a one-week near-miss book for the school: any event that could have hurt someone but did not, one line each, no names. At week's end, read the entries aloud without comment and count them. Bring back the count and one sentence on what the pyramid says about it.

Summative task

Write 160 to 200 words on the following.

A director says, "We have had no accidents for two years, so we are safe." Using the incident pyramid, the leading-versus-lagging distinction and the organizational environment, write the reply, ending with the two measurements you would start this month.

Mini-rubric

Criterion	What a strong response does
Models used (Analyze)	Shows what two quiet years can hide via the pyramid
Indicators (Understand)	Correctly contrasts leading with lagging
Measurements chosen (Apply)	Two concrete, cheap, startable-this-month measures

Criterion	What a strong response does
Clarity and register	Plain sentences for a director; within the word count

Further reading

- World Health Organization. *Occupational health* (health topic):
<https://www.who.int/health-topics/occupational-health>
- Abbasi, H. N., Khan, R., Nasir, I., & Ahmed, W. (2022). A case study of Tariq Road Underpass, Karachi: Environmental analysis. *International Journal of Economic and Environmental Geology*, 12(4).
<https://doi.org/10.46660/ijeeg.Vol12.Iss4.2021.651>
- Harrington, J. M., & Gill, F. S. (Eds.). *Occupational Health*. Blackwell. (guide-recommended book, cited by reference only)

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